

Comments on Proposals: Reduce Minimum Fixed Cost to 75 ADA and Increase k to 1000

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1 Summary of the proposals

The proposal combines two parameter changes intended to strengthen the position of small pools. First, it lowers the minimum fixed cost that can be declared by a pool (`minPoolCost`) from 170 ADA to 75 ADA. This creates a *pull* effect by improving the net rewards that a pool can distribute. Since it is expected that small pools will be the ones reducing the fixed cost, the change would increase small pool attractiveness for delegators.¹ Second, it increases the target number of pools (`stakePoolTargetNum`) from $k = 500$ to $k = 1,000$, which lowers the saturation threshold and causes large pools to become oversaturated more quickly. This creates a *push* effect, as delegators in large pools face stronger incentives to leave and reallocate their stake.

Goals and expected effects. The broader objective is to improve decentralization by making smaller pools more competitive and reducing the relative advantage of large ones. The proposal starts from the view that fixed costs weigh disproportionately on small pools: a larger share of their gross rewards is absorbed by fixed costs, leaving less net reward to distribute to delegators. This problem becomes more severe as the reward pot declines over time while fixed costs remain unchanged. Lowering `minPoolCost` is therefore intended to improve the competitive position of small pools by relaxing this constraint. Allowing small pools to lower their fixed cost may make them more attractive, yet it cannot guarantee a shift in delegation if larger pools maintain higher net rewards. In that case, increasing k acts as a complementary force by pushing delegators out of large pools and making reallocation more likely: once large pools become oversaturated, some delegators are forced to look for alternatives, and smaller pools become more plausible destinations precisely because their attractiveness has already been improved by the reduction in minimum cost.

Open questions. A central question is whether the pull effect generated by the reduction in `minPoolCost` is strong enough on its own, or whether it indeed requires the complementary push effect produced by a higher k . This could be studied, at least in part, by examining the stake dynamics observed after the previous reduction in `minPoolCost`. More generally, the reform should be assessed not only through its direct effects on pool rewards, but also through its impact on

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¹Supporters of the initiative argue that this expectation is motivated by the evidence from the previous reduction in the fixed cost, when many large pools did not adjust their own fixed cost downward. A possible explanation is that these pools remained sufficiently attractive to delegators even while charging a higher fee. Another possibility is that large operators may benefit from a Sybil-type strategy that allows them to collect the fixed fee across multiple pools. This latter mechanism, however, has not been formally proved.

delegation dynamics and operator responses. In particular, one would like to understand whether delegators actually reallocate toward smaller pools when their attractiveness improves, and how incumbent operators—especially large or multi-pool operators—adjust their strategies under the new parameter configuration.

1.1 Summary of takeaways

The main takeaways are as follows.

Summary of Takeaways

- **Impact of Fixed Cost Reductions on Competitiveness.** The incentive structure suggests that cost reductions have a more significant positive impact on the competitiveness of small pools than on that of large ones. Nevertheless, large pools may still remain more attractive for delegators.
- **Effects of a larger k on delegator returns.** After an increase in k , all else equal, delegators in large pools face a significant reduction in net rewards per unit of stake. Delegators in small pools, by contrast, may experience gains or remain largely unaffected.
- **Combined effects on delegator returns.** Holding all else constant, a reduction in `minPoolCost` complements the increase in k by making small pools more attractive to delegators.
- **Ineffectiveness of Fee Wars in Attracting Delegators to Large Pools.** While a "race to the bottom" on fees is a legitimate concern, it remains an ineffective strategy for large pools attempting to counteract the loss of delegator attractiveness caused by an increased k .
- **Effects of a reduction in fixed costs on operators revenues.** A reduction in fixed costs, holding everything else constant—including delegator allocation across pools—reduces pool revenues, with the effect being particularly strong for small pools.
- **Combined impact on operators revenues.** The increase in k largely offsets the negative effect on revenues that a reduction in fixed costs may generate for small pool operators. This is particularly important for those pools that do not receive new delegations.
- **Risk of Reduced Competitiveness for Small Operators.** Even with an increment in k , small pool operators may find fixed cost reductions unprofitable, forcing them to maintain higher costs that diminish their appeal to delegators compared to mid-sized competitors.
- **Strategic Incentives for Medium-Large Pool Operators.** The proposals may incentivize medium-to-large pool operators to optimize returns through strategic pool splitting. Consequently, certain large-scale stakeholders (with many of such pools) could find themselves in a better position after pursuing these optimizations.

2 Analysis

In this part, we aim to partially respond some of the previous open questions. In particular, we complement the existing analysis by considering additional aspects that may help clarify how the proposal affects the different participants in the staking ecosystem (delegators and pool operators).

2.1 Effects over delegators

Reducing the fixed cost increases the net reward that a pool can distribute among its delegators. This is a clear direct effect over *all* pools that decide to reduce their fixed cost. Figure 1 illustrate this benefit comparing the net rewards per unit of stake for different fixed costs and a fixed k . For that comparison, let's first define the net rewards per unit of stake:

$$\frac{f(\sigma_i, p_i) - c_i}{\sigma_i}, \quad \text{where} \quad f(\sigma_i, p_i) = \frac{R}{1 + a_0} \left[\tilde{\sigma}_i + a_0 \tilde{p}_i \frac{\tilde{\sigma}_i - \tilde{p}_i \frac{\tilde{z}_0 - \tilde{\sigma}_i}{\tilde{z}_0}}{\tilde{z}_0} \right],$$

being:

R = reward pot,

a_0 = pledge influence factor,

$z_0 = \frac{T}{k}$ = saturation point,

σ_i = delegations in pool i ,

$\sigma'_i = \min\{\sigma_i, z_0\}$ = delegations in pool i eligible for rewards,

p_i = pledge in pool i ,

$p'_i = \min\{p_i, z_0\}$ = pledge in pool i eligible for rewards,

c_i = fixed cost of pool i ,

$\tilde{\sigma}_i = \frac{\sigma'_i}{T}$,

$\tilde{p}_i = \frac{p'_i}{T}$,

$\tilde{z}_0 = \frac{z_0}{T}$,

T = total supply.

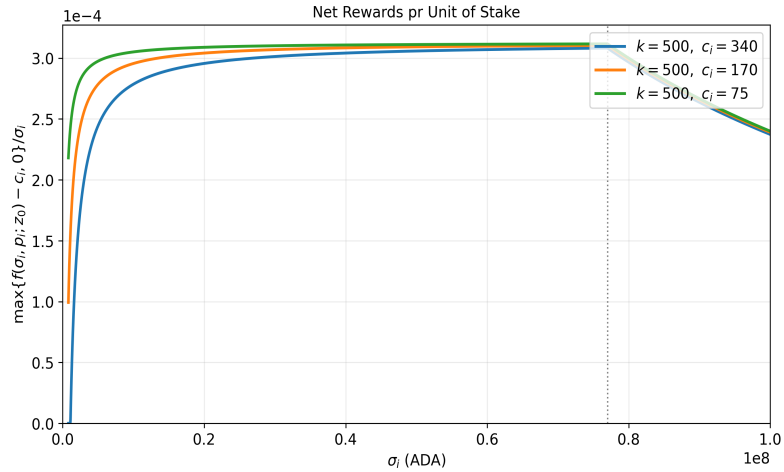


Figure 1: We assume $p = 770K$ ADA, $R = 15.6M$ ADA, $T = 38.5B$ ADA, $a_0 = 0.3$, and $k = 500$.

The previous figure clearly illustrate how the reduction in the fixed cost has a larger impact in smaller pools. However, larger pools still remains more attractive.

Takeaway: Impact of Fixed Cost Reductions on Competitiveness

The incentive structure suggests that cost reductions have a more significant positive impact on the competitiveness of small pools than on that of large ones. Nevertheless, large pools may still remain more attractive for delegators.

A reduction in `minPoolCost` expands the set of options available to pools, which can then choose the response that best suits their situation. By contrast, a change in the parameter k applies mechanically to all pools. Some pools will be negatively affected, while others may benefit, and this is likely to trigger responses from both delegators and operators.

Figure 2 illustrates the effect of an increment in k for different levels of delegation and pledge. In particular, we compare the differences in net rewards when $k = 1000$ versus the benchmark $k = 500$, keeping the fixed cost unchanged.

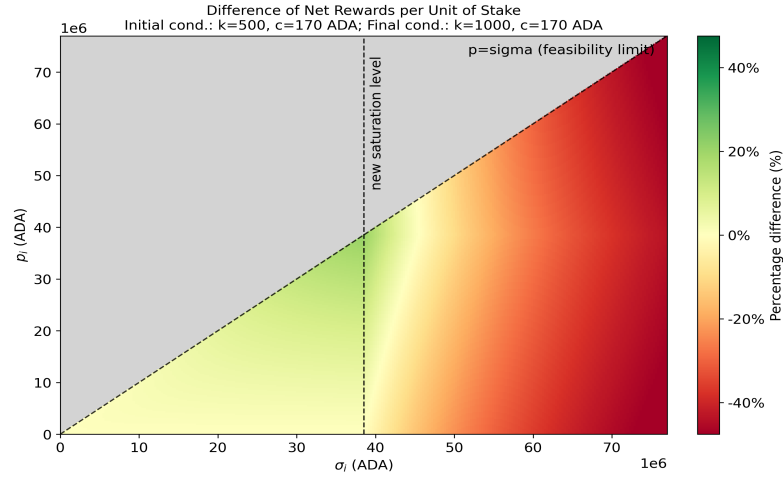


Figure 2: The plot illustrates the net rewards per stake for different combinations of delegation and pledge. We assume $R = 15.6M$ ADA, $T = 38.5B$ ADA, $a_0 = 0.3$, $c_i = 170$ ADA, initial $k = 500$ (implying $z_0 = 77M$ ADA), and final $k = 1000$ (new $z_0 = 38.5M$ ADA).

The figure shows that delegators in small pools (below the new saturation point) benefits (or are almost indifferent) from the large k . As designed in the reward function, this benefit increases with the level of pledge and delegation in the pool. Similar situation face those delegators in pools slightly above the saturation point when the pledge is high enough. Even though they are above the new saturation point, the pools with these parameters are "closer" to the optimal reward level than under $k = 500$. On the other hand, delegators in larger pools will face a reduction in their net rewards, some of them up to near -40% .

Takeaway: Effects of a larger k on delegator returns

After an increase in k , all else equal, delegators in large pools face a significant reduction in net rewards per unit of stake. Delegators in small pools, by contrast, may experience gains or remain largely unaffected.

The fact that delegators in large pools would be strongly affected is likely to induce operator responses along several margins. Highly pledged operators may open additional smaller pools and reallocate their pledge across them, while highly delegated operators may also expand in order to absorb part of the stake displaced by the increase in k . Another possible response, if feasible, is a

reduction in fixed costs aimed at mitigating the negative effect of the increment of k on large pools. Yet this margin may be of limited use for two reasons: (i) lowering fixed costs has only a limited effect on the attractiveness of large pools to delegators, and (ii) small pools may reduce their fixed costs as well. The Figure 1 above, shows these points.

Figure 3 illustrates the case in which all pools reduce their fixed costs as a response to an increment in k , and shows that the gains in net rewards per unit of stake for smaller pools become stronger. Again, the reason is that the positive effect of a reduction in `minPoolCost` on net rewards is much larger for small pools.

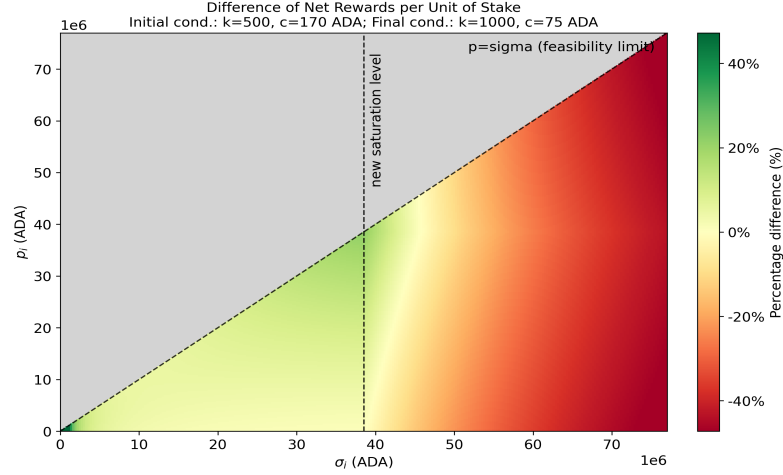


Figure 3: We assume $R = 15.6M$ ADA, $T = 38.5B$ ADA, $a_0 = 0.3$, $c_i = 170$ ADA.

Takeaway: Combined effects on delegator returns

Holding all else constant, a reduction in `minPoolCost` complements the increase in k by making small pools more attractive to delegators.

One critical question arising from the preceding discussion is whether large operators might reduce their fixed costs to offset losses incurred by an increased k , potentially triggering a fee war that marginalizes smaller pools. While this is a valid concern requiring deeper analysis, the figures above suggest that such a strategy is largely ineffective at mitigating the broader impacts of k for large-scale pools. Furthermore, among pools positioned below the new saturation point, a reduction in fixed costs offers a more significant relative advantage to those with low pledge and low delegation than to those operating near the new optimal reward threshold.

Takeaway: Ineffectiveness of Fee Wars in Attracting Delegators to Large Pools

While a "race to the bottom" on fees is a legitimate concern, it remains an ineffective strategy for large pools attempting to counteract the loss of delegator attractiveness caused by an increased k .

The figures above speak only to pool attractiveness from the perspective of delegators. However, a pool may become more attractive to delegators while its profitability deteriorates. In particular, a reduction in `minPoolCost` in isolation may improve the attractiveness of small pools, as discussed above, but at the same time reduce their profitability by lowering the income derived from fixed-cost compensation. We turn to the effects over operators in the next section.

2.2 Effects over Operators

We now consider the effect of the proposal over the pool operators' profits. For that, we use the pool operator revenue function:

$$\begin{cases} c_i + (f(\sigma_i, p_i) - c_i)[m_i + (1 - m_i)\frac{\hat{p}_i}{\sigma_i}], & \text{if } f(\sigma_i, p_i) > c_i, \\ f(\sigma_i, p_i), & \text{otherwise} \end{cases},$$

where we assume that the operator's active pledge is equal to its declared pledge, i.e., $\hat{p}_i = p_i$.

Suppose that there is only a reduction in the `minPoolCost`. Figure 4 shows the effect of this change in the pool operator reward. Notice that small pools (in pledge and delegations) could be negatively affected. This is because the fixed cost plays an important role in operator's revenues. To see this, take

$$\Pi_i = c_i + (f(\sigma_i, p_i) - c_i)[m_i + (1 - m_i)\frac{p_i}{\sigma_i}], \quad \text{if } f(\sigma_i, p_i) > c_i,$$

and define $s = [m_i + (1 - m_i)\frac{p_i}{\sigma_i}]$. Hence, we can rewrite $\Pi_i = sf(\sigma_i, p_i) + (1 - s)c_i$, with $\partial\Pi_i/\partial c_i = (1 - s) \geq 0$. Hence, the operator's profit drops with a lower c_i .

An important caveat is that we are focusing in a situation in which there is no migration of delegators. In the previous subsection, we saw that a lower c_i increases the attractiveness of small pools for delegators. However, in this section we want to explore the effects of the proposals over those pools operators that do not gain or lose delegators.

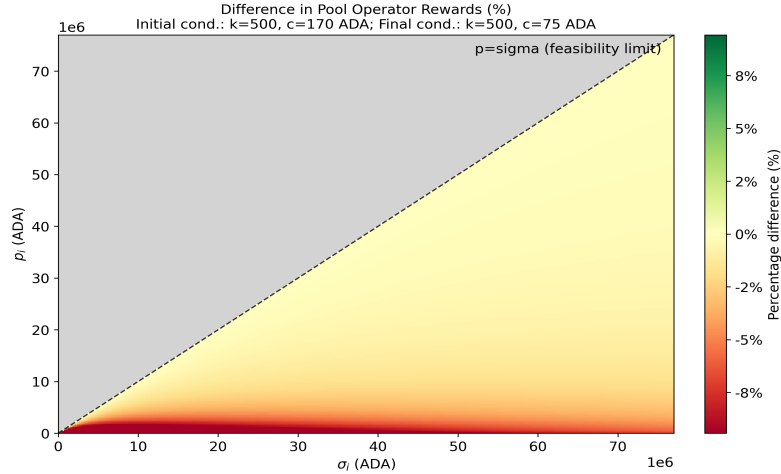


Figure 4: Policy: Only fixed cost reduction. We assume $R = 15.6M$ ADA, $T = 38.5B$ ADA, $a_0 = 0.3$, $m_i = 5\%$.

Takeaway: Effects of a reduction in fixed cost on operators revenues

A reduction in the fixed cost, holding everything else constant—including delegator allocation across pools—reduces pool operator revenues, with the effect being particularly strong for small pools.

The consequence is that very small pool operators will not have room to reduce their fixed costs without losing economic viability.

Next, Figure 5 illustrates the isolated effect of an increase in k on operator revenues. As expected, these results closely mirror those in Figure 2, though the negative impacts are somewhat mitigated by the influence of fixed costs on overall pool profitability.

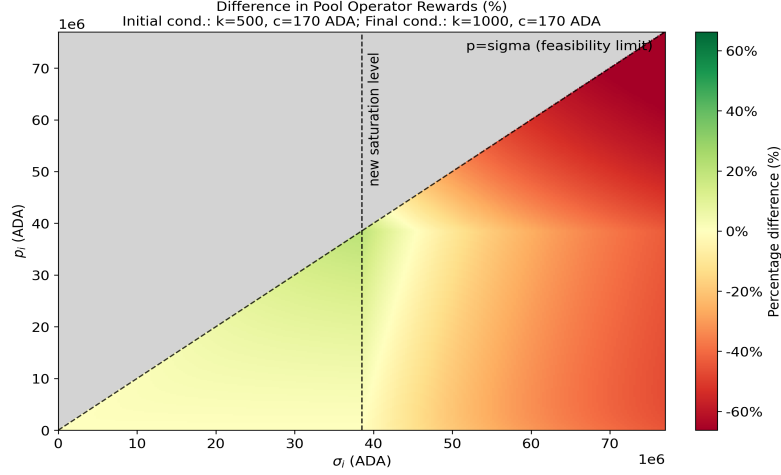


Figure 5: Policy: Only k increment. We assume $R = 15.6M$ ADA, $T = 38.5B$ ADA, $a_0 = 0.3$, .

Taken together, these two effects suggest that the increase in k substantially offsets the negative impact that a reduction in the fixed costs might otherwise impose on small pool operator revenues. This is a critical distinction because, while the "push-pull" design is intended to redistribute delegation from large pools to smaller ones, some smaller pools—particularly those with low pledge amounts—may still struggle to attract new delegates. In such cases, the increment in k serves as a mechanism that may help these pools remain economically viable within the market.

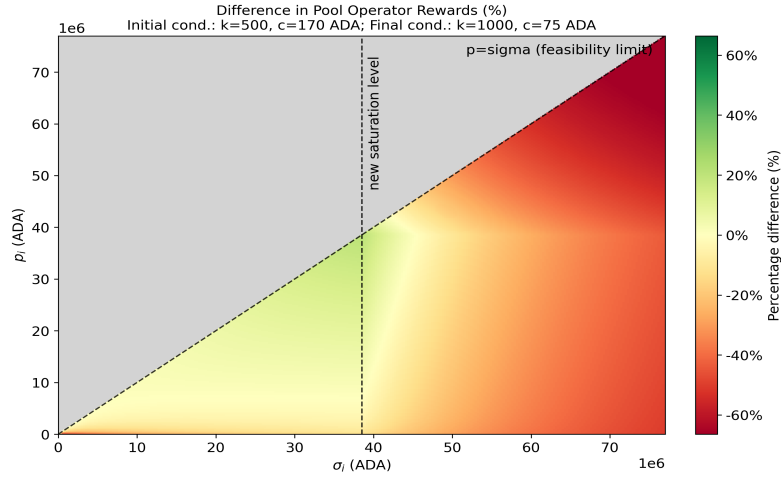


Figure 6: Policy: Both, minPoolCost reduction and k increment. We assume $R = 15.6M$ ADA, $T = 38.5B$ ADA, $a_0 = 0.3$, .

Takeaway: Combined impact on pool profits

The increase in k largely offsets the negative effect on profits that a reduction in the fixed costs may generate for small pool operators. This is particularly important for those that do not receive new delegations.

However, small pool operators may struggle to reduce fixed costs while remaining economically viable. Choosing to maintain higher fees may result in a loss of competitiveness against larger pools.

Takeaway: Risk of Reduced Competitiveness for Small Operators

Even with an increment in k , small pool operators may find fee reductions unprofitable, forcing them to maintain higher costs that diminish their appeal to delegators compared to mid-sized competitors.

In the previous figures, we examined the impact of the changes on pools in the absence of any response, either by operators or by delegators. We now want to understand what an operator could do to improve its situation once the proposals are implemented. Since the set of possible reactions is very large, we focus on the case in which operators follow a Sybil-type behavior. In particular, we consider an operator with one pool, a given level of delegation, and a given pledge, who splits into two pools. For simplicity, we assume that this behavior is followed only by operators whose pools become oversaturated after the proposals are implemented. When calculating and comparing the rewards, we consider the total reward of the pool operator rather than the reward at the individual pool level. Figure 7 shows that large pool operators can offset the negative effects generated by the proposals over their revenues if they are able to split and achieve a full reallocation of their current delegators across the new two pools.

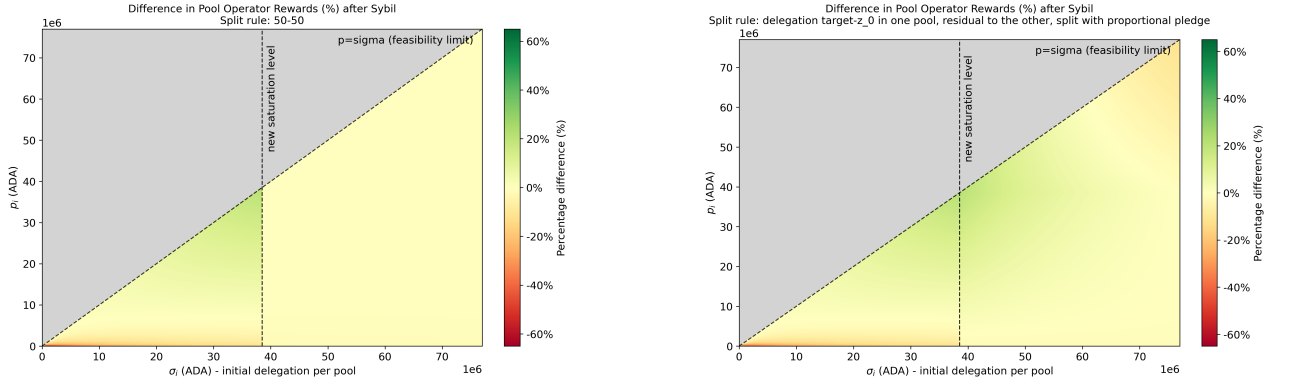


Figure 7: Difference of operators reward after they split into 2 identical pools (left), and saturating with delegation one pool while keeping proportional pledge (right). We assume $R = 15.6M$ ADA, $T = 38.5B$ ADA, $a_0 = 0.3$, $m_i = 5\%$. Initial conditions: $k = 500$, $c_i = 170$ ADA; Final conditions: $k = 1000$, $c_i = 75$ ADA.

For instance, operators of medium-to-large pools (e.g., 50M stake and 30M pledge) may improve their revenues by splitting into two distinct pools with identical pledges, provided they can fully saturate one of them with delegation (as seen in the figure on the right). Conversely, an operator of a pool already characterized by high pledge and high delegation (the top-right quadrant in both figures) would be worse off by splitting in this manner; the light orange hue in the right-hand figure indicates a decrease in revenues. This same operator would, however, remain indifferent if the split were executed as a 50-50 distribution of both pledge and delegation across two pools.

These findings suggest that operators of medium-to-large pools—a category that includes certain “whales”—may benefit from the current proposals, since the proposals push them to optimize their pools. This observation naturally raises the question of whether other strategies, beyond those illustrated here, might exist that could similarly benefit the operators of the largest pools.

Notice that the previous findings are based on the strong assumption that these operators do not lose delegators and manage to convince them to reallocate in their new pool. This could be the case of those whales that manages the stakes of their users, e.g., CeXs.

Takeaway: Strategic Incentives for Medium-Large Pool Operators

The proposals may incentivize medium-to-large pool operators to optimize returns through strategic pool splitting. Consequently, certain large-scale stakeholders (with many of such pools) could find themselves in a better position after pursuing these optimizations.